

III. Results

To evaluate the effectiveness of the Transport of Intensity Equation (TIE) reconstruction, the processed intensity derivatives, recovered phase distributions, and corresponding surface height maps are presented in Figures 2–25. These results demonstrate how patterns generated by the display code propagate through the reconstruction process and appear in the resulting 3D profiles. Notably, the reconstructed surfaces exhibit apparent depth variations that closely follow the imposed intensity patterns, even in cases where no physical height variation is expected. This behavior suggests that the TIE algorithm is highly sensitive to spatial intensity gradients introduced by the display. Because TIE relates axial intensity variations to phase, structured illumination can be misinterpreted as phase curvature, leading to artificial depth features in the reconstructed surface. This highlights a key limitation of TIE-based reconstruction under non-uniform illumination conditions.

To emulate spatial light modulation, an Arduino-controlled LCD was used to generate programmable intensity patterns, analogous to beam shaping in a spatial light modulator (SLM). In this implementation, patterns were created using C++ code to systematically vary the spatial distribution of light intensity when illuminated by a laser source. Figure 1 presents representative examples of these generated patterns prior to their use in the TIE reconstruction process. The patterns exhibit variations in shape, size, and spatial frequency, providing controlled test conditions for evaluating the response of the reconstruction algorithm. The goal is to determine whether the TIE reconstruction can accurately reproduce these imposed intensity variations as corresponding features in the recovered phase and 3D surface profiles.

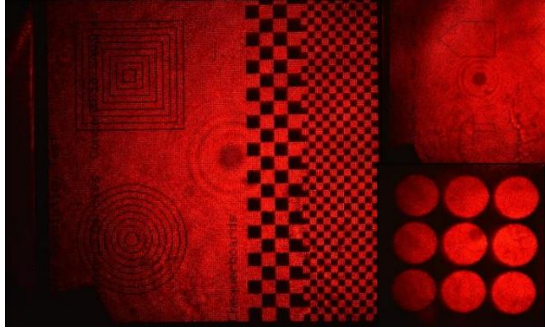


Figure 1 shows examples of spatial light modulation patterns generated using the Arduino-controlled LCD.

March 2026 results

For the initial TIE reconstruction, the LCD was used in an unmodulated state, with no applied Arduino control signal, effectively representing the SLM in an “off” condition. In this case, the LCD surface itself including inherent defects and structural features served as the source of intensity variation. Shown in figures 2-7.

The reconstruction reveals several consistent features of the physical LCD. In particular, a prominent defect located near the bottom of the screen, likely a scratch or dent, is clearly reproduced in the 3D surface profile. Additionally, diagonal line structures observed on the LCD surface are also visible in the reconstructed image, particularly in the upper region, indicating that the TIE method is capable of capturing real surface characteristics under these conditions. However, the overall reconstruction exhibits significant irregularities, appearing lumpy and wave-like rather than smooth. This behavior is consistent with earlier trials in which similar wave-like artifacts were observed. These distortions are likely due to noise in the system, including environmental factors such as air fluctuations, particle interactions, and instability in illumination when the SLM is not actively controlled. To mitigate these effects, noise reduction techniques were implemented in the MATLAB reconstruction code; however, residual artifacts remain, suggesting that noise continues to influence the recovered phase and height profiles. This

suggests that while TIE can recover true surface features, its accuracy is strongly affected by noise when intensity modulation is not well controlled.

Broken LCD (SLM Off)

Trial 1

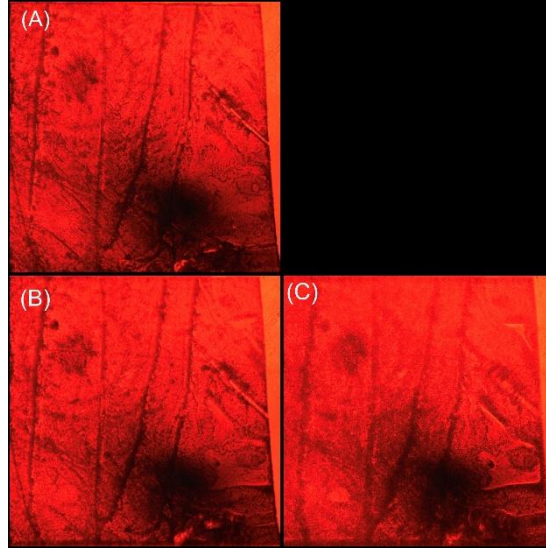


Figure 2 shows an Lcd scrap from peeling. (A) Primary image (B) 5mm ahead of I_0 (c) 5mm behind I_0

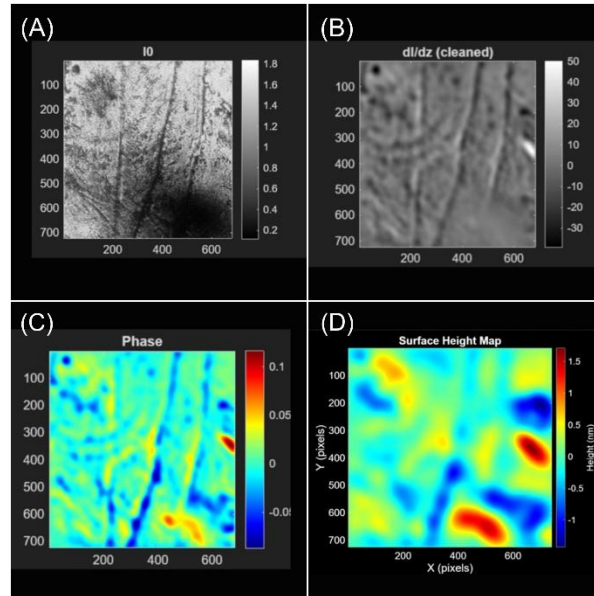


Figure 3 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

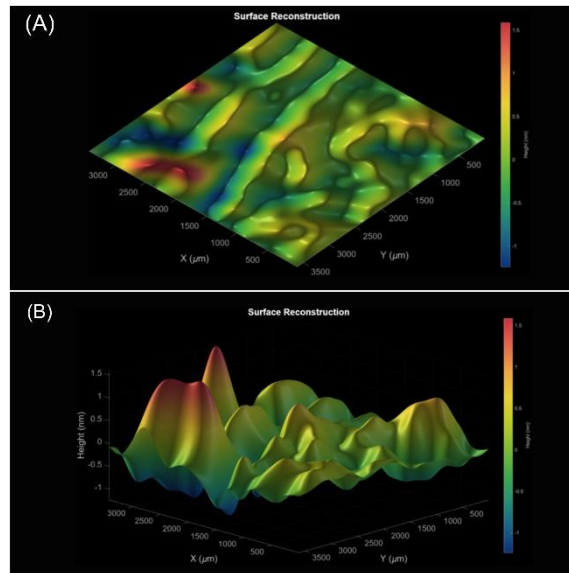


Figure 4 shows the 3d reconstruction of the lcd slm turned off and shows characteristics of the damage. (A) top view (B) side view

Trial 2

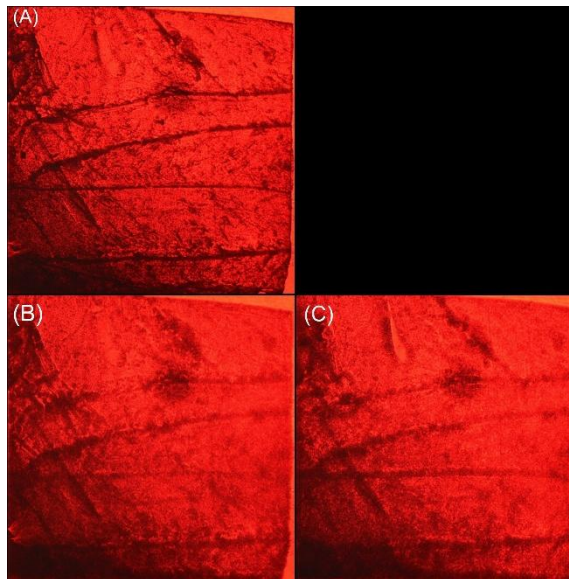


Figure 5 shows an Lcd scrap from peeling. (A) Primary image (B) 5mm ahead of I0 (c) 5mm behind I0

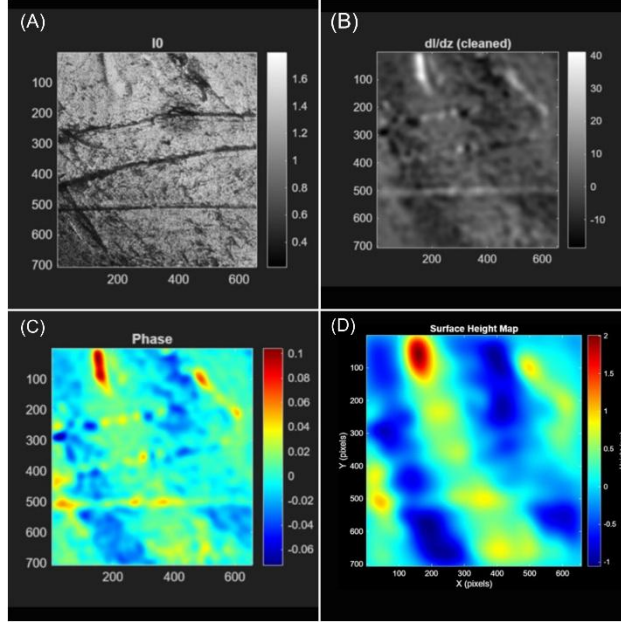


Figure 6 shows Figure 3 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

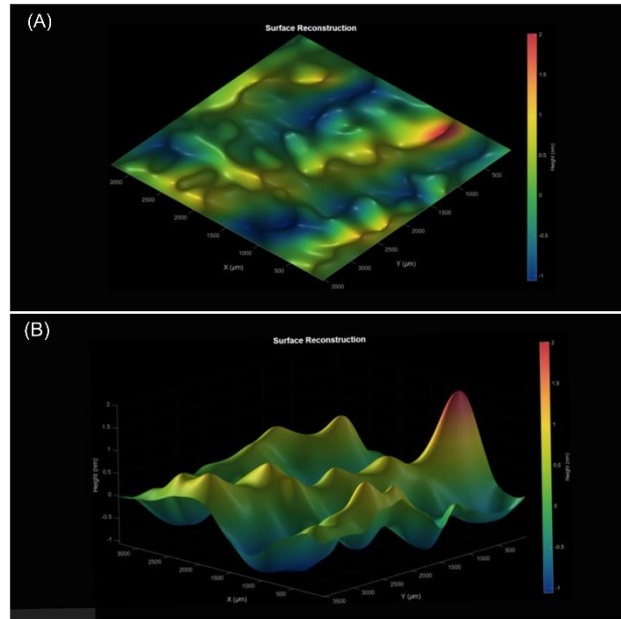


Figure 7 shows the 3d reconstruction of the lcd slm turned off and shows characteristics of the damage. (A) top view (B) side view

The TIE reconstructions were next performed with the SLM actively modulated using Arduino-generated patterns. Figures 8- 25 present TIE reconstructions obtained under patterned illumination for both intact and damaged LCD conditions. Despite differences in the specific patterns applied, all reconstructions exhibit consistent behavior: the imposed intensity patterns introduce structured variations in the recovered phase and surface height maps. In the case of the intact LCD (Figures 8-10), periodic features corresponding to the projected patterns are clearly visible, even in regions where no physical surface variation is present. This indicates that the TIE algorithm is interpreting intensity modulation as phase structure, resulting in artificial surface features.

For the damaged LCD (Figures 11-25), these effects are further amplified. Existing defects interact with the imposed patterns, producing exaggerated peaks and valleys in the reconstructed surface. As a result, defects appear significantly more pronounced compared to the unmodulated case, demonstrating that structured illumination enhances both real and artificial features.

Additionally, variations in pattern spatial frequency influence the appearance of the reconstruction. Higher-frequency patterns produce more localized and rapidly varying features, while lower-frequency patterns result in smoother, wave-like structures. These results confirm that the observed depth variations are not solely due to physical surface geometry but are strongly influenced by intensity gradients introduced by the display.

Normal LCD (SLM On)

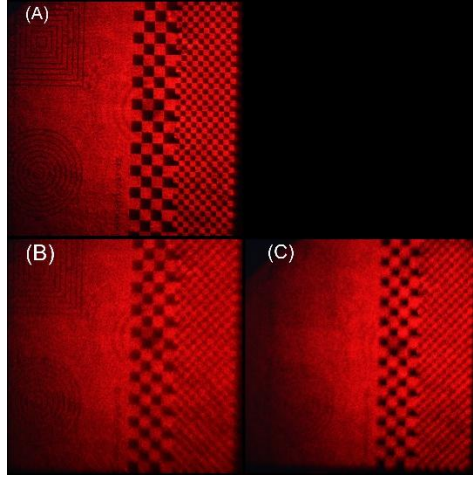


Figure 8 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z).

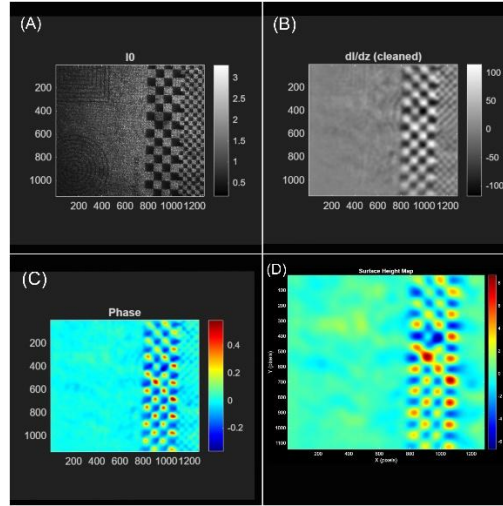


Figure 9 (A) shows the in-focus intensity I_0 , serving as the reference image with surface texture and illumination variations. (B) shows the axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) shows the recovered phase distribution, highlighting optical path variations in the LCD. Planar tilt was removed to isolate structural features, with minor noise present in low-intensity regions due to small I_0 values. (D) presents the corresponding two-dimensional height map.

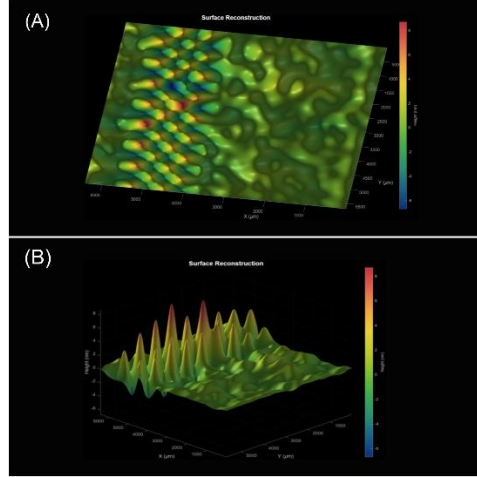


Figure 10 displays the three-dimensional surface reconstruction of a normal LCD with the SLM displaying a pattern, shown (B) side and (A) top views. Peaks and valleys correspond to texture features in the original intensity images.

Broken LCD (SLM On)

Checkerboard circles and squares

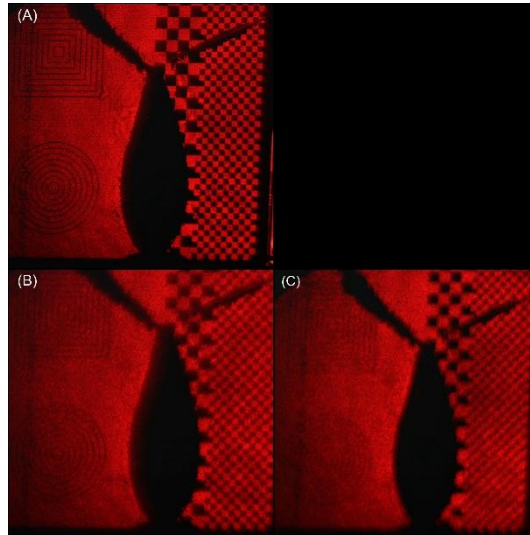


Figure 11 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a checkerboard pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

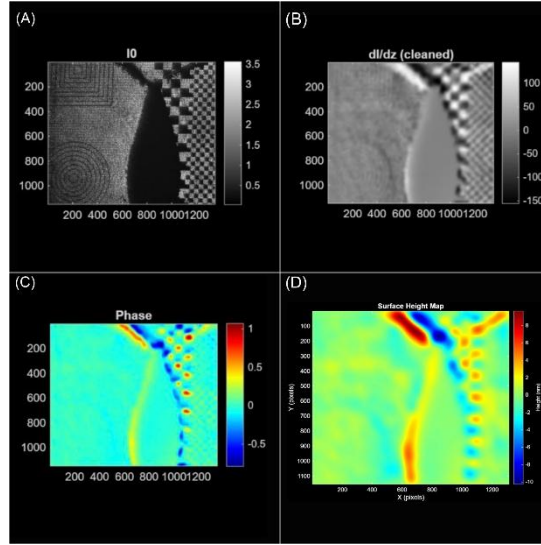


Figure 12 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

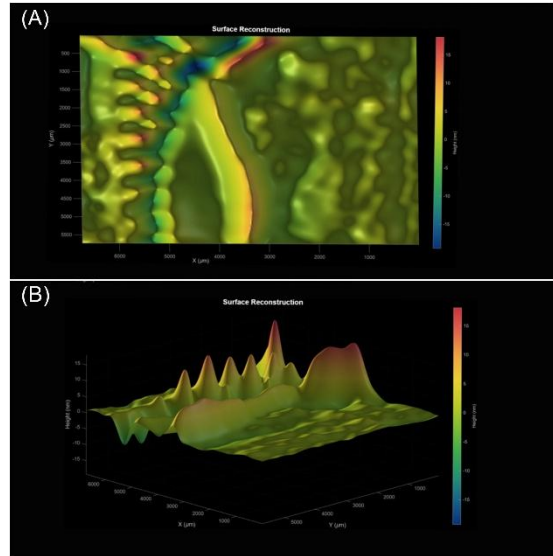


Figure 13 shows the 3d reconstruction of the checkered damaged lcd. The damages show to be more intense than the checkered patterns. (A) top view (B) Side view

Squares on the right side

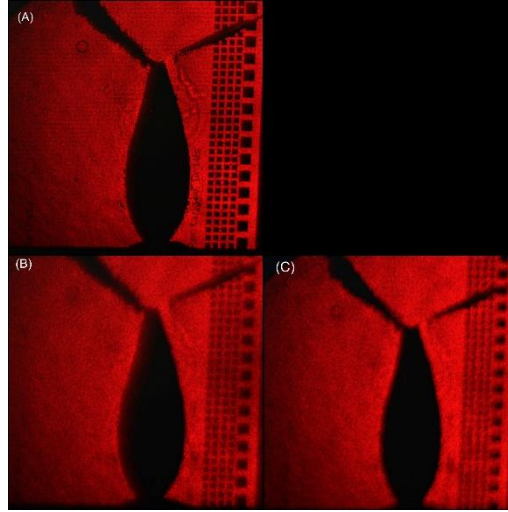


Figure 14 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Squares on the right-side pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (−z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction.

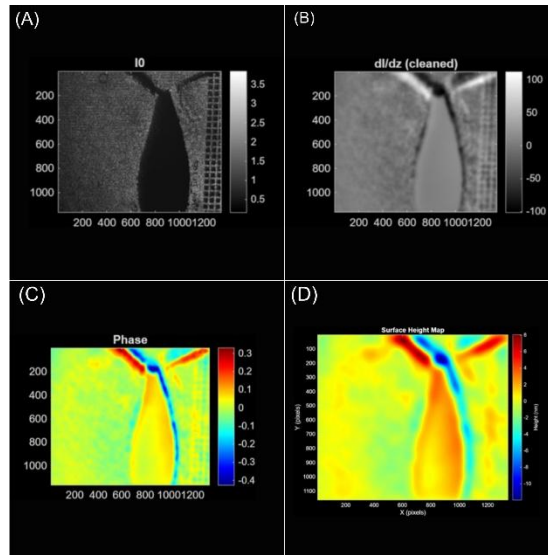


Figure 15 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

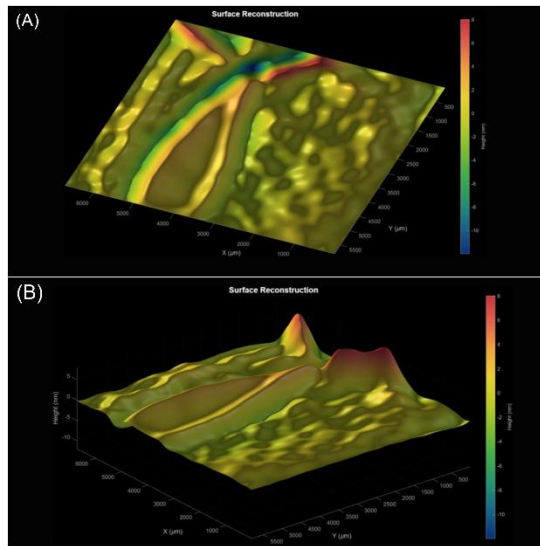


Figure 16 shows the 3d reconstruction of the checkered damaged lcd. The damages show to be more intense than the checkered patterns. (A) top view (B) Side view

April 2026 Results

Additional patterns were generated to replicate common spatial light modulator (SLM) outputs, including diffractive structures such as concentric circles and line gratings. These patterns were used to further evaluate the response of the TIE reconstruction under structured illumination.

The resulting reconstructions are consistent with the corresponding in-focus intensity images I_0 , with the imposed patterns clearly reflected in the recovered phase and surface height maps. This agreement indicates that the TIE algorithm effectively captures the spatial distribution of intensity variations introduced by the display.

In the case of the damaged LCD, the defect remains clearly visible in the reconstruction and is further enhanced by the imposed patterns. This reinforces previous observations that TIE reconstruction is highly sensitive to both physical surface features and intensity modulation, with defects appearing more pronounced under structured illumination.

Broken LCD (SLM On)

Stripes and Interference circle waves

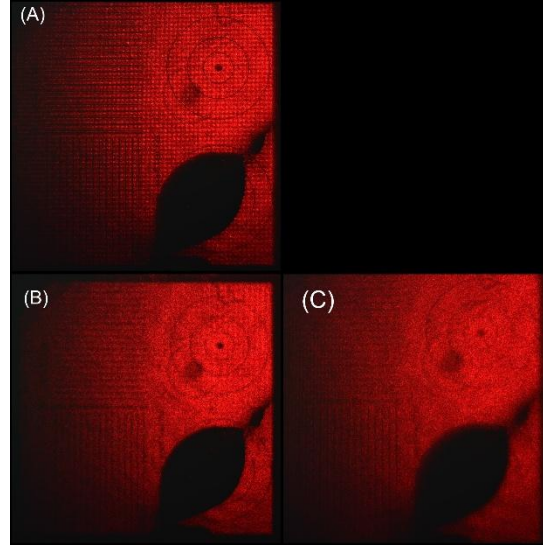


Figure 17 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying Stripes and interference circle waves pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

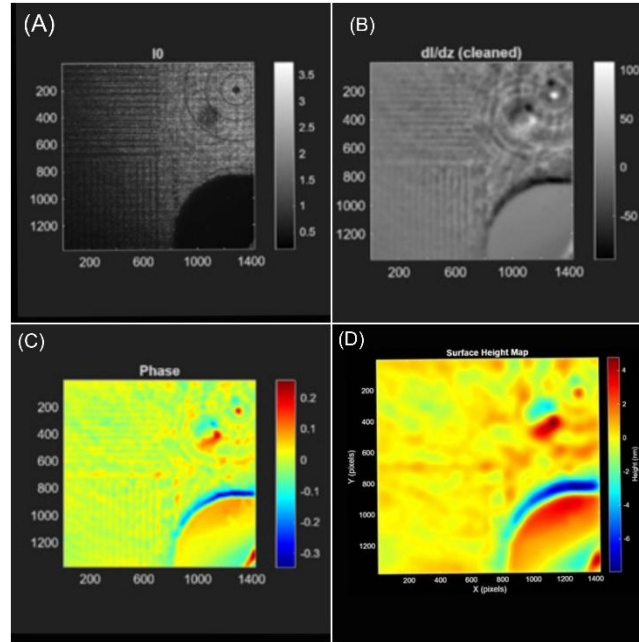


Figure 18 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

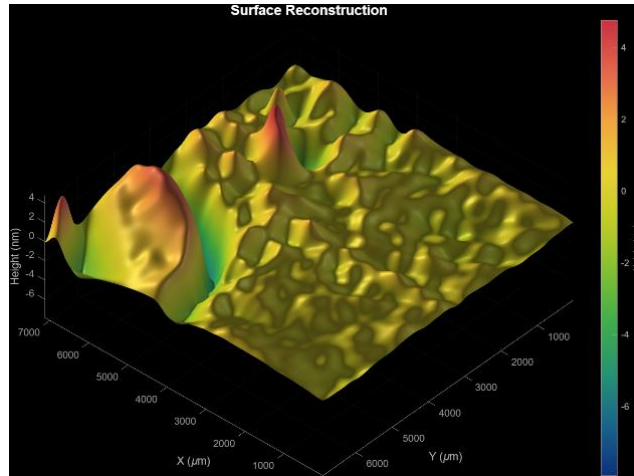


Figure 19 shows the three-dimensional reconstruction of the projected pattern. The damaged region of the LCD appears more pronounced than the imposed checkerboard pattern, indicating that surface defects produce stronger intensity variations and dominate the TIE reconstruction.

Stripes

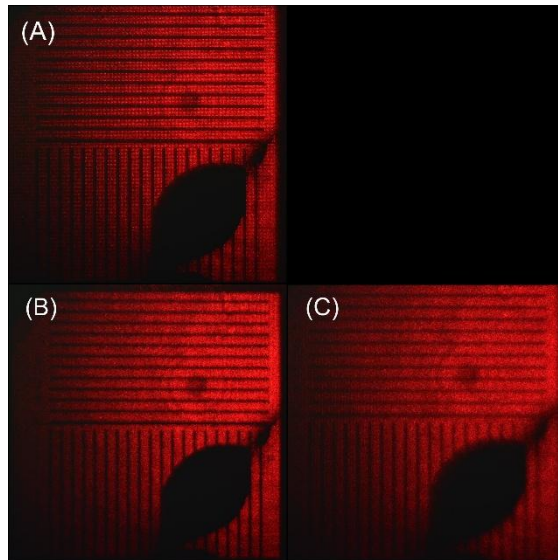


Figure 20 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying a Stripes pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

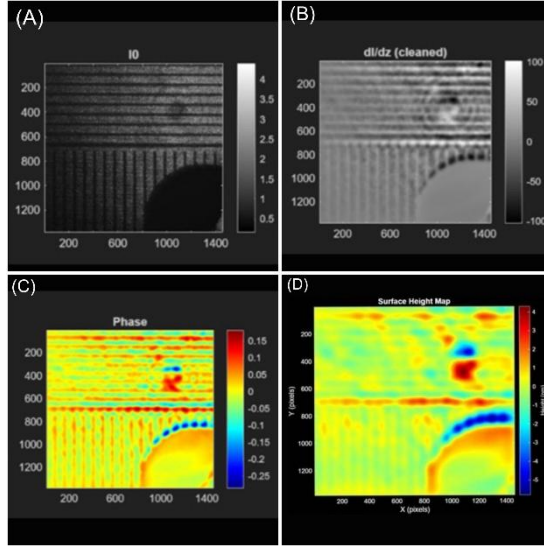


Figure 21 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I/\partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

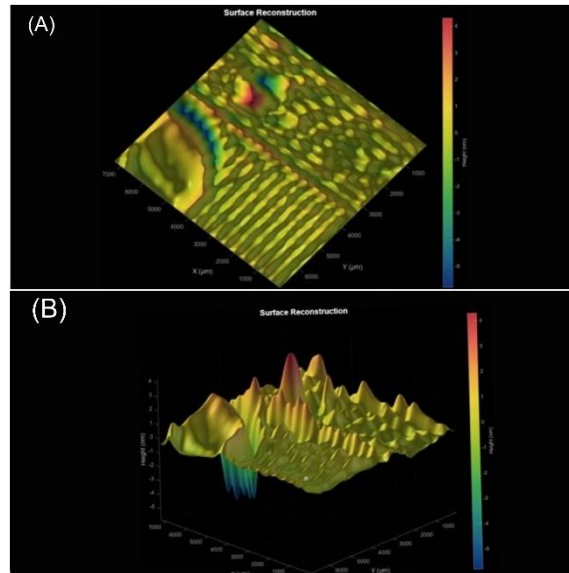


Figure 22 shows the three-dimensional reconstruction of the stripe pattern. The reconstructed features are consistent with the reference image I_0 , with the stripe structure clearly preserved. Additionally, characteristics of the damaged LCD, including surface irregularities, are visible in the reconstruction.

Interference circle waves

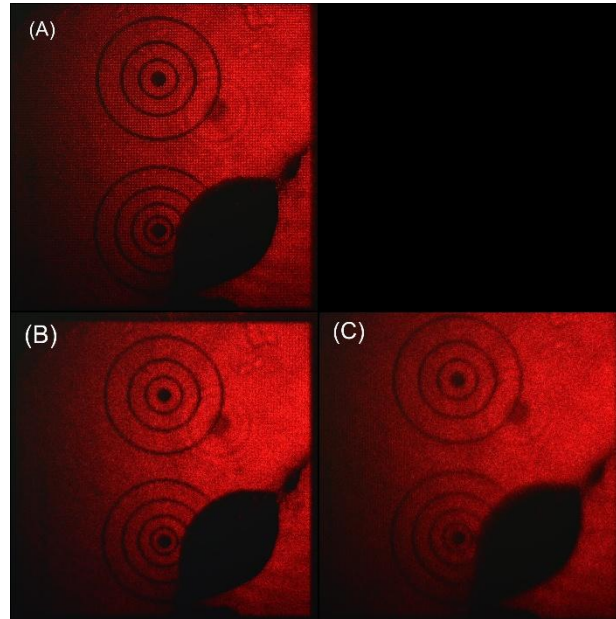


Figure 23 shows Intensity images of an Arduino-controlled LCD used as a spatial light modulator (SLM) displaying an interference circle wave pattern on a damaged screen. (A) In-focus intensity image I_0 . (B) Intensity image acquired 5 mm ahead of the focal plane (+z). (C) Intensity image acquired 5 mm behind the focal plane (-z). The damaged region of the LCD is visible in all images and interacts with the imposed pattern, providing the input data required for TIE reconstruction

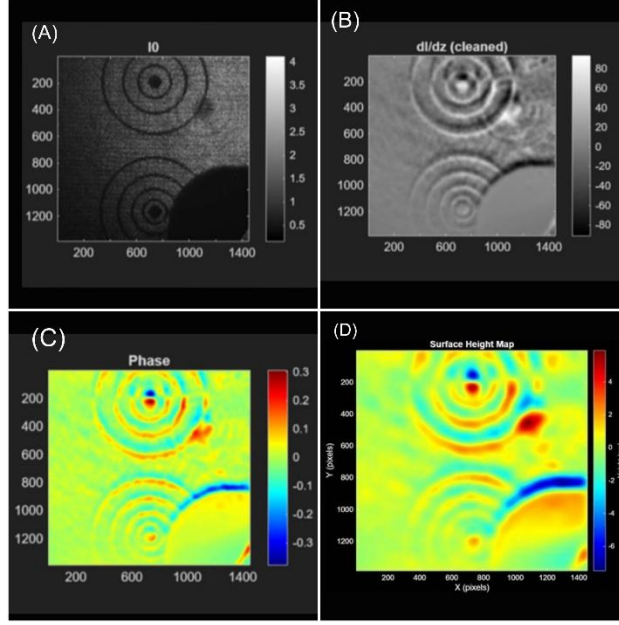


Figure 24 shows (A) In-focus intensity image I_0 , showing inherent surface texture and illumination variations. (B) Axial intensity derivative $\partial I / \partial z$, highlighting edges and structural features. (C) Recovered phase distribution, representing optical path variations across the surface. (D) Corresponding surface height map, where localized peaks and valleys reflect surface irregularities of the LCD.

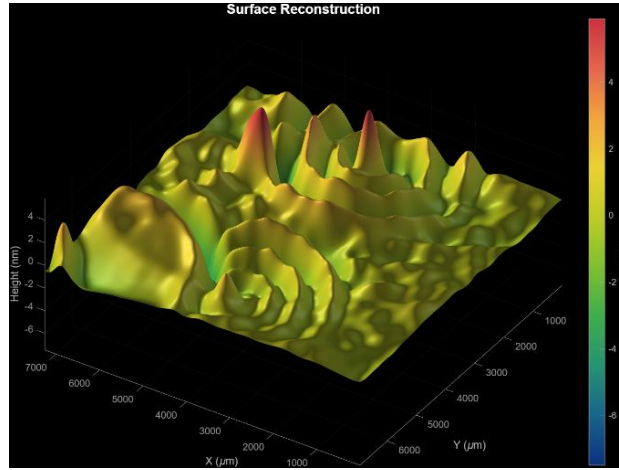


Figure 25 shows the three-dimensional reconstruction of the circular pattern. The reconstructed features closely match the reference image I_0 , with the concentric ring structure clearly preserved. Surface defects in the damaged LCD are also evident, demonstrating that both the imposed pattern and inherent screen irregularities contribute to the reconstructed surface.

IV. Discussions and Conclusions

Overall, the TIE reconstruction successfully recovered phase information from intensity-only measurements and produced a quantitative three-dimensional surface map of the LCD material. Despite the presence of minor low-frequency curvature and residual noise, the reconstructed structures correspond well with features observed in the raw intensity images, supporting the validity of the method.

The results demonstrate that phase nonuniformities and surface irregularities in the LCD can be effectively characterized without the need for interferometric techniques, highlighting the potential of the Transport of Intensity Equation as a low-cost and accessible tool for optical surface analysis and spatial light modulator evaluation.

However, the findings also reveal an important limitation: the TIE reconstruction is highly sensitive to spatial intensity variations. Under structured illumination, imposed patterns can be interpreted as phase variations, leading to artificial or exaggerated surface features in the reconstructed height maps. This indicates that care must be taken when applying TIE in systems with non-uniform or dynamically modulated illumination, as the recovered phase may not solely represent true physical geometry.

V. Future Works

To further validate this work, additional experimental conditions will be explored to better assess the accuracy and performance of the low-cost LCD-based SLM system. While the current results demonstrate strong capability in capturing intensity variations and reconstructing surface features, further investigation is required to fully characterize the limitations of the method.

Future studies will examine the SLM under a wider range of operating conditions, including uniform (no-pattern) illumination while the device is actively driven. This will help isolate the intrinsic phase response of the LCD and distinguish it from pattern-induced effects.

Additional efforts will focus on improving noise reduction techniques and system stability to minimize wave-like artifacts observed in the reconstruction. Expanding these experimental conditions will provide a more comprehensive understanding of the optical behavior of low-cost SLM systems and enhance the reliability of TIE-based phase reconstruction.